

## Technology Stack

|                      |                                                         |
|----------------------|---------------------------------------------------------|
| <b>Date</b>          | 2 October 2026                                          |
| <b>Team ID</b>       | 12                                                      |
| <b>Project Name</b>  | ComicCraft - AI Comic Story Creator using Gemini Models |
| <b>Maximum Marks</b> | 2 Marks                                                 |

### Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

### Technology Stack Details

| S.No | Architecture Component / Layer | Technology Chosen                                                                                                                         | Justification / Purpose                                                                                   |
|------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 1    | Frontend / Client-Side         | HTML, CSS, Jinja2 templates; Bangers and Poppins fonts                                                                                    | Server-rendered pages keep the app simple; responsive glass-style form with pill options                  |
| 2    | Backend / Server-Side          | Python 3.14, FastAPI, Uvicorn                                                                                                             | Fast to build, typed, automatic API docs at /docs; team knows Python                                      |
| 3    | Database / Data Storage        | No database. Files in static/panels (PNG) and static/exports (PDF)                                                                        | A comic is generated and downloaded in one session; nothing needs to be queried later                     |
| 4    | Cloud / Hosting / Deployment   | Runs locally with Uvicorn (Apple Silicon MPS, CUDA or CPU)                                                                                | Avoids GPU hosting cost; Docker and a cloud GPU are listed in the future plan                             |
| 5    | Version Control & CI/CD        | Git and GitHub; pytest (24 tests) run locally                                                                                             | Team collaboration through pull requests; CI is planned                                                   |
| 6    | Third-Party APIs / Other Tools | Google Gemini (google-generativeai); Stable Diffusion v1.5 via diffusers, PyTorch, transformers, accelerate; fpdf2; Pillow; python-dotenv | Gemini free tier for text; local model for images at no per-image cost; fpdf2 for PDF; dotenv for secrets |